

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3 FT	(a)			In clockwise direction: cell / battery (1) <u>variable</u> resistor / rheostat (1) voltmeter (1) don't accept voltmeter / voltage ammeter (1) don't accept ampmeter	4			4		4
	(b)	(i)		100 mA identified anyway			1	1		1
		(ii)	(I)	0.05 [A]		1		1	1	1
			(II)	Substitution: $\frac{1}{0.05}$ (1) ecf = 20 [Ω] (1)	1	1		2	2	2
		(iii)		All 8 points correct $\pm <1$ small square division (2) 7 points correct $\pm <1$ small square division (1) 1-6 points correct $\pm <1$ small square division (0) Correct curve of best fit starting between 0.4 - 0.6 V up to 1.0 V (1) Don't accept thick, double, wispy or disjointed curves		3		3	3	3
		(iv)		Reading taken from candidate's graph ± 1 mA		1		1	1	1
		(v)		The current increases as the voltage increases / positive correlation (1) at an increasing rate / each increase gets bigger / use of numbers e.g. current increased by 5 [mA] between 0.4 and 0.6 [V] and increased by 30 [mA] between 0.8 and 1 [V] (1) Don't accept in a non-proportional way / the increase gets steeper / non-linear		2		2		2
		(vi)		Diode / LED Don't accept circuit symbol			1	1		1
				Question 3 total	5	8	2	15	7	15